

Towards Model-Agnostic Cooperative Perception

Junjie Wang¹, Tomas Nordström^{1,2}

¹Department of Applied Physics and Electronics, Umeå University

²RISE Research Institutes of Sweden

Abstract—Existing cooperative perception systems often require joint training among participating agents. However, this assumption clashes with practical deployments where agents belong to diverse entities, each employing unique models and performing their own downstream tasks. Sharing model details for joint training is often hard to achieve since the concerns of intellectual property (IP) leakage and the centralized training may conflict with the downstream task of each agent. To address these challenges, we introduce IMCP, a robust Intermediate Model-Agnostic Cooperative Perception framework. IMCP enables universal agent fusion without the need for joint training or model sharing. Each agent undergoes independent initial training, followed by a cooperative fine-tuning stage where the feature encoder of each agent remains frozen. To effectively fuse features from diverse domains, we incorporate parameter-efficient hierarchical feature adaptation layers that map features into a common representation space. Furthermore, deformable attention is employed to selectively aggregate multiple Bird's-Eye View (BEV) features of varying sizes. Extensive experiments on two real-world cooperative perception datasets demonstrate that IMCP achieves comparable performance to existing joint training methods. Code is available at <https://github.com/JesseWong333/IMCP>.

Index Terms—Cooperative Perception, Autonomous driving

I. INTRODUCTION

Cooperative perception [1, 2] involves the fusion of sensory data from multiple autonomous agents to construct a more comprehensive and precise understanding of the surrounding environment. Unlike single-agent perception systems, cooperative perception leverages the collective intelligence of multiple agents to enhance perception performance and mitigate challenges such as occlusions, sensor noise, and ambiguities. This cooperative approach enhances the robustness and reliability of perception systems, making them applicable across various domains, including autonomous driving and robotics. Recent studies [3–8] have demonstrated the significant advantages of cooperative perception in 3D object detection and semantic segmentation tasks compared to single-agent perception systems. Moreover, these studies have addressed practical challenges, including communication bandwidth limitations [4, 9, 10], localization errors [8, 11], latency issues [12–15], and the presence of heterogeneous devices [6, 16, 17].

Despite the numerous advantages offered by cooperative perception over single-agent perception systems, current methods often rely on joint training of all participating agents to adapt to cooperative operation, as shown in Fig. 1. However, practical deployments frequently involve agents from diverse entities utilizing distinct models and configurations. Joint training in such contexts presents significant drawbacks. Firstly,

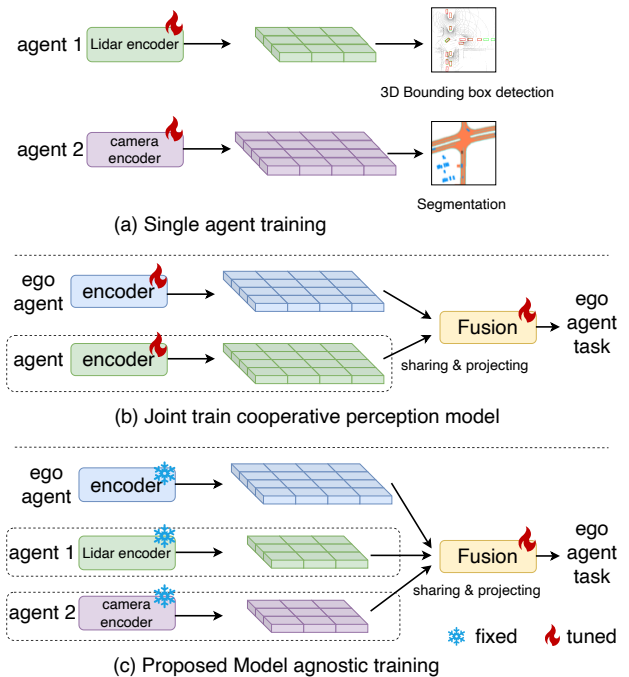


Fig. 1: Comparison of training methods. (a) In conventional single-agent autonomous driving, agents from different entities are trained independently and perform their tasks without any communication. (b) In existing cooperative perception frameworks, a unified perception neural network is jointly trained, and individual modules are subsequently deployed to each agent. (c) In contrast, our approach leverages existing, independently trained single-agent models. The fusion module is fine-tuned to adapt to the specific task of the ego agent while preserving the integrity of the single-agent models.

considering the large number of agents and the continuous new agents joining the cooperation, it's hard to train just once. Secondly, these models are inherently heterogeneous, employing diverse model architectures (e.g., Point-Pillars [18], SECOND [19], BEVFormer [20]), configurations (e.g., varying grid sizes), and sensor modalities (e.g., lidar, camera). They may be trained for different downstream tasks, such as 3D object detection, segmentation, tracking, or end-to-end self-driving. It's hard to consider the heterogeneity in joint training, especially for heterogeneous downstream tasks. Thirdly, the entities have already built powerful, carefully optimized single-agent neural models considering their equipped sensors and computation capacity. Training a model dedicated to cooperative perception

might worsen when only one agent exists. Lastly, this joint training might require the details of these models, which might cause intellectual property (IP) leakage of different entities. We argue that intermediate model-agnostic cooperative perception is an ideal solution for these challenges. In the model-agnostic cooperative perception system, from the perspective of an ego agent (the agent that receives information from other agents), all collaborating agents essentially act as *black-box* models, where only intermediate outputs are accessible. It's inherently heterogeneous, can hide the model details, and can directly use the large number of existing single-agent model. Existing model-agnostic cooperative perception considered result-level fusion [21], while it's impossible to fuse models with different downstream tasks. Also, result-level fusion usually shows inferior performance than intermediate fusion [6, 7].

In this paper, we propose the first Intermediate Model-Agnostic Cooperative Perception (IMCP) framework, specifically designed to address the practical challenges inherent in this emerging field. To mitigate the domain gap arising from diverse feature representations received from other agents, we incorporate feature adaptation layers. These layers aim to map the received features into the domain of the ego agent's representations. This design is inspired by prior work [22] which demonstrated that representations learned by deep neural networks with similar architectures, even with different inputs, exhibit a degree of inherent similarity. This similarity can often be effectively mapped using a single linear layer. Based on this observation, we hypothesize that representations of the same scenario, generated by heterogeneous models deployed on different agents, still also share a significant level of similarity. Furthermore, considering that intermediate features from the same model family tend to be even more similar, we propose using hierarchical feature adaptation layers. These layers are designed to minimize the number of parameters requiring tuning, thereby enhancing computational efficiency. To effectively fuse features of varying dimensions, we introduce a multi-level and multi-agent fusion module that leverages deformable attention [23] to integrate intermediate feature maps. This approach allows each agent to participate in cooperative tasks by solely tuning the attention weights and sampling offsets within the deformable attention mechanism.

We have done extensive experiments on two real-world cooperative perception datasets, V2V4real [24] and Dair-V2X [25], to evaluate the effectiveness of IMCP. Our results demonstrate that IMCP facilitates efficient cooperative perception when cooperating agents employ diverse model architectures, modalities, tasks, and configurations. In summary, the contributions of this work include:

- We formulate the model-agnostic challenge in cooperative perception for the first time and present an intermediate model-agnostic cooperative perception framework¹.
- We propose a parameter-efficient hierarchical feature adaptation method, which largely reduces the finetuning

memory footprints and storage space compared to existing joint training methods.

- we empirically demonstrate that trained neural networks with different architectures, tasks, and modalities that represent the same scene can be fused and improve the overall performance.
- Our evaluations demonstrate IMCP can obtain comparable performance with existing joint-training methods and achieve parameter efficiency when adapting to a new agent.

II. RELATED WORKS

Our work addresses the challenges of model agnostic cooperative perception. This approach leverages hierarchical feature adaptation layers to effectively integrate single-agent models into a cooperative system. This research intersects with several key areas, including cooperative perception and model stitching techniques.

A. Cooperative perception

Cooperative perception allows multi-agents to share perceived information through the communication network, significantly improving the perception capabilities over single-agent perception. Both simulated datasets [27, 28] and real-world datasets [24, 25, 29] have been developed to support algorithm development. Researchers have made significant progress in addressing challenges inherent to cooperative perception systems, such as communication bandwidth limitations [4, 9, 10], localization errors [8, 11], latency issues [12–15], and the presence of heterogeneous devices [6, 16, 17]. Additionally, some research focuses on camera-only methods [5] for 3D object detection and semantic occupancy prediction [30]. However, a common characteristic of these methods is the requirement for joint training of all models, which, in our opinion, presents several significant drawbacks as previously discussed. Still, existing model-agnostic cooperative perception considered result-level fusion [21], which focuses on recalibrating the 3D detection boxes from multiple agents. HEAL [31] considers incorporating new agents into an existing framework. This approach adopts a strategy of training only the encoder of the newly added agent while keeping the fusion model fixed. In contrast, this work explores intermediate model-agnostic fusion, enabling cooperation among agents with diverse model architectures, modalities, tasks, and configurations.

B. Representations matching and model stitching

The concept of representation similarity has been explored in previous works to analyze the compatibility of different network representations [22, 32]. Lenc et al. [32] pioneered the concept of model stitching by demonstrating that early layers of one trained network could be effectively stitched with the final layers of another through a simple linear layer, with minimal performance degradation. This finding suggests a degree of equivalence between representations learned at different stages within trained networks. Yamini et al. [22]

¹A concurrent work [26] was brought to our attention during review.

further extended this concept by demonstrating that networks with distinct architectures or training strategies could also be stitched together with minimal performance impact. Notably, Pan et al. [33] applied the concept of stitching to transformer models, which holds promise for efficient neural architecture search. Merullo et al. [34] demonstrate that image representations extracted from vision models can be transferred as continuous prompts for frozen language models by training a simple linear projection. These theoretical analyses and practical applications demonstrate that even a simple linear layer can effectively transfer features across different domains. Building on this insight, our hierarchical feature adaptation layers employ a similarly lightweight transformation, yet extend beyond the capabilities of a single linear layer.

III. PROBLEM FORMULATION

Cooperative perception is a technique enabling multiple agents to collectively perceive a scene by exchanging perceived information through wireless communication, e.g. V2X communication [35]. Generally, there are three types of cooperation in the previous literature: early fusion, intermediate fusion, and late fusion. Early fusion can be communication-intensive and may not fully leverage the computational capabilities of individual agents. Late fusion, although model-agnostic, necessitates identical tasks among agents and often exhibits inferior performance [6, 7, 21]. Given these limitations, this paper exclusively focuses on intermediate fusion. Consider a scenario with N agents in the same scene, each agent $n \in \{1 \dots N\}$ senses the environment using its sensors, and processes the data $\mathbf{X}_n$ captured to obtain a feature map $\mathbf{F}_n$. Subsequently, these processed features, along with the agent's pose, are broadcasted through a communication network. The feature extraction method remains consistent with the single-agent approach [18, 36, 37]. Features received from other agents are then projected to the ego-agent's coordinate system, a process denoted as f_{proj} . Finally, the ego agent aggregates all features to generate a fused feature map $\hat{\mathbf{F}}_{ego}$ at the ego's coordinate system, denoted as f_{agg} . In the model-agnostic scenario, all cooperating agents function as *black-box* models. This implies that the encoder of each agent, denoted as $f_{encoder}^n$, is fixed and cannot be modified by the ego agent. The downstream outputs of cooperating agents, denoted $\mathbf{Y}_n$, need not be directly related to the primary task objective of the ego agent. The fused feature map is then fed into a detection head $f_{head}(\cdot)$ to get the final result $\mathbf{Y}_{ego}^t$. The cooperative perception tasks can be formulated as follows:

$$\mathbf{F}_n = f_{encoder}^n(\mathbf{X}_n), n \in \{1 \dots N\}, \quad (1)$$

$$\mathbf{Y}_n = f_{head}(\mathbf{F}_n), \quad (2)$$

$$\mathbf{F}'_n = f_{proj}(\mathbf{F}_n), \quad (3)$$

$$\hat{\mathbf{F}}_{ego} = f_{agg}(\mathbf{F}_{ego}, \mathbf{F}'_{n, n \neq ego}), \quad (4)$$

$$\mathbf{Y}_{ego} = f_{head}(\hat{\mathbf{F}}_{ego}). \quad (5)$$

Here, $f_{agg}(\cdot)$ is the aggregation operator.

IV. INTERMEDIATE MODEL-AGNOSTIC COOPERATIVE PERCEPTION FRAMEWORK

In this section, we present our robust intermediate model-agnostic cooperative perception framework, which can effectively fuse feature maps from other independently trained agents.

A. Overall Architecture

As illustrated in Fig. 2, our framework follows an encoder-decoder architecture [20, 23], which consists of a feature encoder, a model-agnostic fusion decoder with deformable cross/self-attention layers, and a task-specific head.

For each agent n , the feature encoder processes the raw data and produces BEV features $\mathbf{F}_n \in \mathbb{R}^{H_n \times W_n \times C_n}$. To further adapt the features for the non-ego agent, an additional parameter-efficient feature adaptation layer was incorporated. Similar to [20], we first initialize a set of grid-shaped learnable parameters $Q \in \mathbb{R}^{bev_h \times bev_w \times bev_c}$ as BEV queries, where bev_h , bev_w are the spatial dimensions of the BEV plane, each grid $q_{p=(x,y)} \in \mathbb{R}^{1 \times bev_c}$ within the BEV queries representing the feature within a spatial grid at location $p = (x, y)$. Additionally, positional embeddings are added to BEV queries Q before being sent to the fusion decoder. Our fusion process in Eq. 4 is based on deformable attention, which can be formulated as:

$$\text{DeformAttn}(q, \{p_n\}_{n=1}^N, \{\mathbf{F}_n\}_{n=1}^N) = \sum_{i=1}^{N_{\text{head}}} \mathcal{W}_i \left[\sum_{n=1}^N \sum_{j=1}^{N_{\text{key}}} \mathcal{A}_{ijn} \cdot \mathcal{W}'_i \mathbf{F}_n(p_n + \Delta p_{nij}) \right], \quad (6)$$

where N_{head} is the total number of attention heads, N_{key} is the total number of sampled keys for each head. $\mathcal{W}_i$ and $\mathcal{W}'_i$ are learnable weights. $\mathcal{A}_{ijn} \in [0, 1]$ is the predicted attention weight, and is normalized by $\sum_{n=1}^N \sum_{j=1}^{N_{\text{key}}} \mathcal{A}_{ijn} = 1$. We use $\Delta p_{nij} \in \mathbb{R}^2$ to denote the predicted offsets relative to the reference point p_n .

B. Feature adaptation

Existing literature [22, 33, 34] has demonstrated the efficacy of even simple linear layers in transferring features across diverse domains. This observation suggests that feature maps extracted from neural networks, despite being trained for distinct tasks or modalities, can exhibit substantial similarity, particularly when derived from representations within the same scene. While linear layers have proven effective in certain scenarios, our investigations indicate that slightly deeper architectures, comprising a few additional layers, can significantly enhance performance. Furthermore, we observed varying degrees of feature similarity between different agents. For instance, models belonging to the same family generally demonstrated higher levels of feature similarity. To improve parameter efficiency during feature adaptation and effectively leverage both inter-agent and intra-agent representational similarities, we propose a hierarchical adaptation layer, as shown in Fig. 3. This design enables the selective reuse of layer

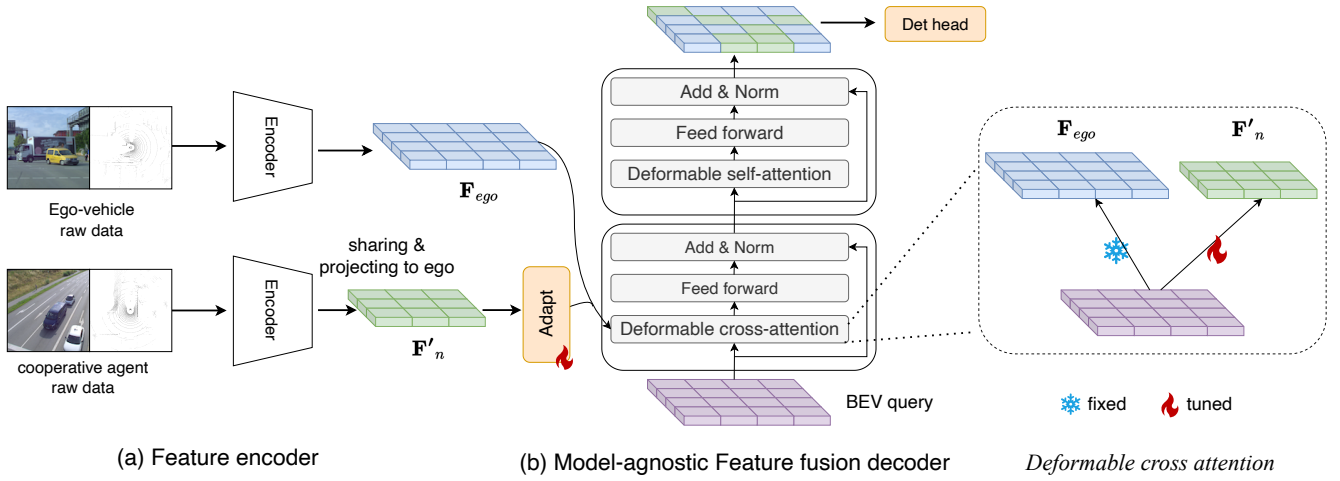


Fig. 2: **Overall architecture.** (a) The feature encoder transforms the raw data into BEV features. The parameters of the encoder are optimized by its own task and fixed during the cooperative training stage. (b) To aggregate multiple Bird's-Eye View (BEV) features, we employ deformable attention. For non-ego agents, a lightweight, tunable feature adaptor is incorporated. Additionally, the attention weights and sampling offsets prediction layer associated with the non-ego agent layer can also be tuned.

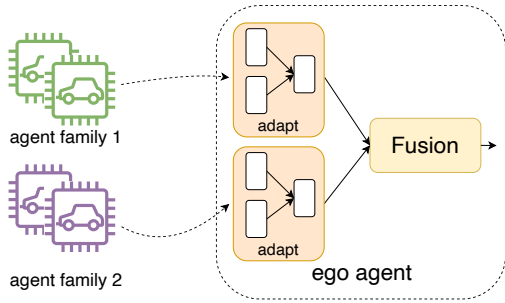


Fig. 3: Illustration of the hierarchical feature adaptation layer.

components, utilizing a greater number of parameters for models with lower similarity and a smaller subset for models with higher similarity. We replace with the Eq. 3 as:

$$\mathbf{F}'_n = f_{adapt}(f_{proj}(\mathbf{F}_n)), \quad (7)$$

C. Multiple Agent Feature Fusion

Following the feature adaptation, a crucial step involves fusing these features from multiple agents. We identified two primary challenges in designing an effective fusion module: 1) handling feature heterogeneity: In a model-agnostic setting, agents may produce features of varying sizes or scales. Our fusion mechanism must accommodate this diversity without imposing strict constraints on agent architectures. 2) ensuring robustness to agent dropout. In cooperative perception scenarios, communication links can be unreliable. The fusion module should function efficiently even when some agents are unavailable or their contributions are lost. To address these challenges, we introduce deformable attention [23] as the core of our fusion mechanism. Deformable attention exhibits inher-

ent flexibility, enabling seamless integration of features with diverse spatial extents. Specifically, for each new cooperating agent, two linear layers need to be tuned:

- Predicting the pre-normalized attention weights $\mathcal{A}'_{inj}$.
- Predicting the sampling offsets for deformable attention Δp_{nij} .

Each layer has only $n_{dim} \times 2$ parameters. A unique learnable embedding is assigned to each agent-feature level pair, encoding the agent's model and the specific feature level. This approach allows for dynamic and adaptive feature fusion, effectively addressing the challenges of heterogeneity and robustness in cooperative perception systems.

D. Training and inference

Fig. 4 illustrates a real-world model-agnostic cooperative perception scenario. The proposed cooperative perception framework follows a two-stage training process. In the initial stage, individual agents, affiliated with distinct entities, are trained independently. This stage involves meticulous optimization of single-agent neural models, carefully considering the specific sensors and computational capabilities of each agent. Subsequently, in the cooperative fine-tuning stage, only a limited subset of parameters necessitates fine-tuning, namely the feature adaptation layer and the two linear layers as previously discussed. All data remains localized during the cooperative fine-tuning stage, and the training process is executed locally. During the inference stage, upon successful confirmation of an agent's integration into the cooperative perception system, the ego agent dynamically loads the corresponding per-agent associated parameters (i.e., the parameters fine-tuned during the cooperative fine-tuning stage) and subsequently performs cooperative perception.

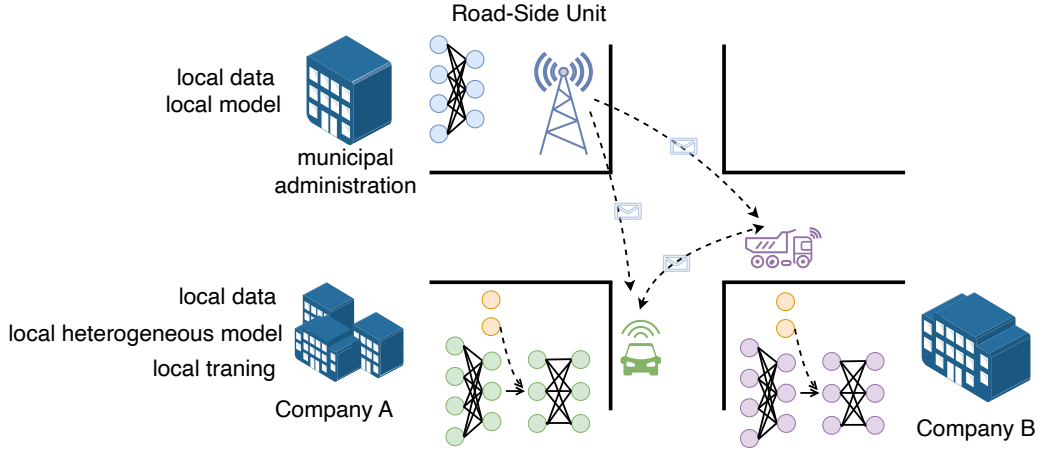


Fig. 4: Illustration of real-world model-agnostic cooperative perception scenario.

V. EXPERIMENTS

A. Datasets

V2V4real [24]. V2V4real is a real-world vehicle-to-vehicle cooperative perception dataset featuring two vehicles within each scene. The dataset covers a driving area of 410 km, comprising 20K lidar frames. As the RGB data is currently unavailable, our analysis focuses exclusively on lidar-based tasks. For the segmentation task, 3D bounding boxes are projected onto the bird's-eye-view plane to generate ground truth targets.

Dair-V2X [25]. Dair-V2X is another real-world cooperative perception dataset, capturing data from one ego vehicle and one roadside unit within each frame. In total, Dair-V2X provides a collection of 6,617 frames for cooperative perception, each captured at a rate of 10 Hz with 3D annotations. We adopted the 3d bounding box annotations as in [8], which complemented the missing label to cover 360-degree detection.

B. Implementation details

For the V2V4real dataset, the agent with id 0 is selected as ego agent. For the Dair-V2X dataset, the vehicle side is always selected as an ego agent. We use two methods, PointPillar [18] and SECOND [19] for the lidar modality (denoted as L_p & L_s), one method Lift, Splat, Shoot (Lss) [37] for camera modality (denoted as C). We use superscript seg to denote the segmentation task. Since our method can be easily used for multiple-scale fusion, in here we consider the fusion with different scale features. For example, for PointPillar we use downsampled features with 2, 4, 8 times and 4 times for the SECOND method. A summary of the setting of these methods can be found at Tab. I.

For V2V4real data, we follow the setting of the benchmark [24], at the training stage we use the range of $\pm 140.8m \times \pm 38.4m$ and we only evaluate objects within the range of $\pm 100m \times \pm 40m$. The detection range is $\pm 140m \times \pm 40m$ for the Dair-V2X dataset.

TABLE I: Basic agent setting for model-agnostic cooperative perception

Agent	L_p	L_s	C	L_p^{seg}
Modality	Lidar	Lidar	Camera	Lidar
Downsampling	2x,4x,8x	4x	2x,4x,8x	4x
Method	PointPillar	SECOND	PointPillar	Lss
Task	Detection	Detection	Detection	Segmentation

For the ego agent, we stacked 4 deformable cross-attention layers and 2 deformable self-attention layers as the model-agnostic feature fusion decoder. Each layer has 8 heads and 8 sampling points. All collaborators in our experiments follow the 1-round communication setting [8], in which each agent shares the pose and feature map simultaneously. We used the same anchor-based detector head and losses for 3D object detection task as in [6, 8, 9, 27], with an initial learning rate of $2e^{-3}$ to train the detection model. For Dair-V2X dataset, we train each model for 20 epochs and 60 for V2V4real dataset. We use the weighted cross-entropy with a learning rate of $2e^{-4}$ for the segmentation task. Background and foreground weighting values were set at 0.005 and 1, respectively. The 3D object detection task is based on the Average Precision (AP) metric [38] at the specified Intersection-over-Union (IoU) thresholds of 0.5 and 0.7. The segmentation metric is based on the IOU score at the BEV grid level.

C. Evaluation results

Tab. II shows the performance of the non-ego agents, evaluated based on their own detection ranges. The results indicate that both PointPillar and SECOND with lidar modality can achieve comparable performance, while the camera modality performs significantly worse. Tab. III shows the cooperative 3D detection performance on the V2V4real and Dair-V2X datasets, in comparison to benchmark including single agent, V2XViT [6], openv2v [27], where2comm [9] and HEAL [31]. The HEAL results are based on our own re-implementation of the backward alignment method as de-

TABLE II: Performance of the non-ego agent.

dataset	V2V4real		Dair-V2X	
Metric	AP@0.5	AP@0.7	AP@0.5	AP@0.7
L_p	84.7	57.4	86.6	80.0
L_s	90.3	74.9	87.2	80.8
C	-	-	36.7	10.1
Metric	IOU			
L_p^{seg}	50.1		56.2	

TABLE III: Comparison of 3D detection performance.

Dataset	V2V4real		Dair-V2X		
Metric	AP@0.5	AP@0.7	AP@0.5	AP@0.7	#para. tuned (M)
Single (L_p)	58.1	38.1	68.6	59.9	-
V2X-ViT	64.9	36.9	70.4	53.1	18.2
Openv2v	64.5	34.3	73.3	55.3	8.69
Where2comm	62.5	34.2	79.9	65.9	12.9
HEAL(back. align.)	64.9	32.8	79.4	66.3	8.41
$L_p + L_p$	67.4	32.1	80.2	67.9	0.79
$L_p + L_s$	70.0	37.3	80.1	67.6	0.42
$L_p + L_p^{seg}$	68.7	33.5	79.8	67.4	1.68
$L_p + C$	-	-	73.3	59.6	1.14

scribed in the original HEAL paper. Notably, our method demonstrates competitive results, often surpassing existing approaches. When cooperating with different agent types, e.g. varying methods, tasks, or modalities, our proposed method consistently achieves significant performance gains compared to single-agent detection. However, we observed one exception: AP@0.7 does not surpass single-agent performance on the V2V4real dataset, which is likely attributable to pose errors in the dataset. While cooperation with camera modalities yields lower performance than with lidar agents, this is expected given the generally lower perception performance of camera-based systems. Furthermore, our method exhibits remarkable parameter efficiency. Adaptation to a new agent requires only a small fraction of parameters (often less than 2%), highlighting the efficiency of our approach.

Tab. IV presents the performance of cooperating with models from the same family, including models trained with different initializations, datasets (utilizing 80% of the data for training L_s), and backbones. Notably, our results indicate that comparable performance can be achieved by reusing some adaptation layers and only fine-tuning one single linear layer, suggesting a hierarchical design that significantly reduces the storage space required for agent-specific parameters. Still, as discussed in [22], there is no definitive metric to measure the similarity between the feature maps. Generally, models belonging to the same family exhibit substantial similarities.

D. Ablation

We also compare the proposed method with a widely employed parameter-efficient fine-tuning (PEFT) technique, namely low-rank adaptation (LoRA). In these experiments,

TABLE IV: Performance of L_p cooperating with Model family of L_s on Dair-V2X dataset

Model variety	AP@0.5	AP@0.7	#para. tuned (M)
Baseline	80.1	67.6	0.42
Diff. init.	80.4	67.6	0.12
Less data	80.1	67.1	0.12
Diff. backbone	80.4	67.6	0.12

TABLE V: Ablation study for L_P+L_P scenario on Dair-V2X.

Metric	AP@0.5	AP@0.7	#para. tuned (M)
LoRA (r=16)	78.6	63.6	0.53
LoRA (r=32)	79.0	65.4	0.70
LoRA (r=128)	79.2	66.1	1.73
Tune fusion module	80.0	66.3	5.00
Ours	80.0	67.8	1.24

we focus on tuning the fusion module without using an adaptation layer. From Tab. V, we see that while increasing the number of LoRA tuning parameters gradually improves the performance, it still falls short of matching the performance achieved by tuning the entire fusion module. In contrast, our method achieves comparable performance while tuning significantly fewer parameters. This finding suggests that, in the context of multi-agent fusion, introducing a limited number of adaptations is more efficient than tuning the entire fusion model.

Table VI demonstrates the impact of the number of adaptation layers on model performance. Notably, even with a single adaptation layer, the method achieves a respectable level of performance. Further enhancement can be observed by adding one or two additional layers, quickly converging to performance levels comparable to tuning the entire fusion module.

E. Qualitative evaluations

Visualization of detection results. Fig. 5 and Fig. 6 visualize the detection results on the V2X4Real dataset and Dair-V2X dataset, in comparison to single-agent detection results and HEAL [31]. We see that single-agent detection exhibits limitations arising from the restricted field of view and sensor constraints, leading to challenges in handling

TABLE VI: Effect of the Number of adaptation layer on Dair-V2X dataset

	# layers	1	2	3	4
L_P+L_s	AP@0.5	78.7	79.6	80.1	80.1
	AP@0.7	65.6	66.4	67.6	66.7

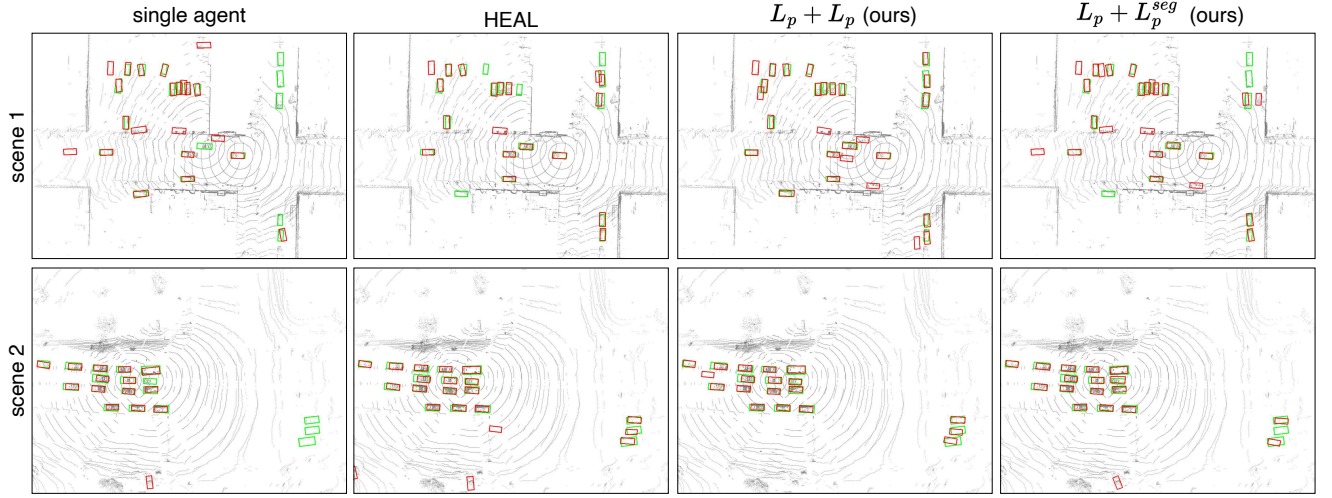


Fig. 5: Visualization of the detection results on V2X4real datasets. We show the detection results of single agent, HEAL, L_P+L_P , and $L_P+L_P^{seg}$. Red boxes: detected, Green boxes: ground-truth

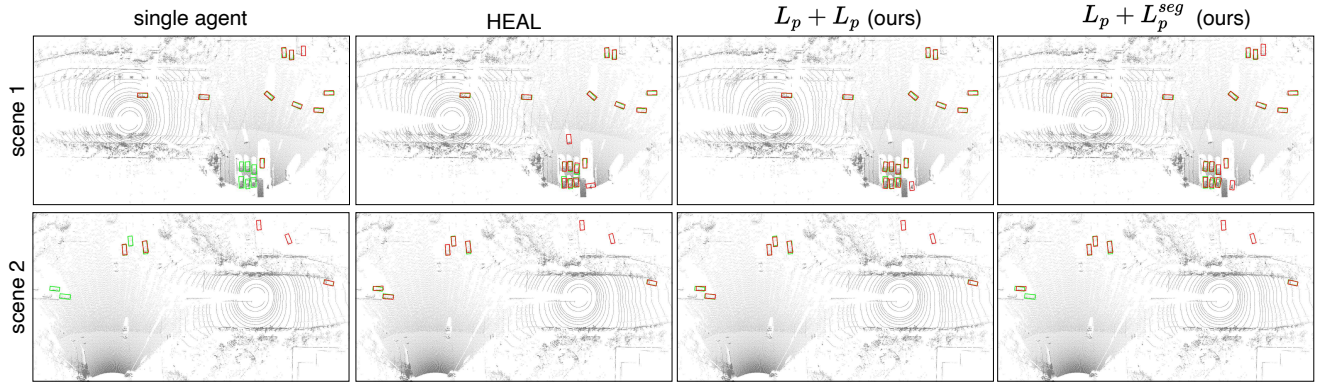


Fig. 6: Visualization of the detection results on DAIR-V2X datasets. We show the detection results of single agent, HEAL, L_P+L_P , and $L_P+L_P^{seg}$. Red boxes: detected, Green boxes: ground-truth

occlusions. For instance, in the V2X4Real dataset, single-agent detection frequently misses objects in densely populated urban environments where buildings and other structures obstruct visibility. Both the HEAL, the L_P+L_P , and $L_P+L_P^{seg}$ cooperation method extend the view range beyond the single-agent's limitations. Compared to the existing methods, our proposed framework achieves comparable performance while exhibiting flexibility in accommodating various neural network architectures.

Visualization of feature map adaptation. Fig. 7 and Fig. 8 visualize the feature maps in the L_P+L_P , and $L_P+L_P^{seg}$ fusion scenario. The visualizations demonstrate that a limited number of adaptation layers can significantly alter the activation patterns of the feature maps. We observed two typical patterns in the feature adaptation process: 1) The feature map $\mathbf{F}'_n$ exhibiting incorrect activations or values close to zero were transformed to appropriate activation levels, as shown in the top rows of Fig. 7 and Fig. 8. 2) The feature map $\mathbf{F}'_n$ with

incorrect activations were transformed to zero, as shown in the bottom rows of Fig. 7 and Fig. 8. These visualization results indicate that the proposed method is capable of effectively adapting the feature maps to the target feature space, ensuring compatibility and enhancing the fusion process.

VI. LIMITATION AND DISCUSSION

The proposed framework aims to achieve model-agnostic cooperative perception by introducing feature adaptation training. However, several challenges remain in realizing a fully model-agnostic cooperative perception system. First, our solution assumes the existence of an explicit spatial feature representation, where each grid corresponds to a localized feature space in the environment. This assumption may not hold for certain perception algorithms that lack such explicit spatial representations, thereby limiting their integration into the proposed framework. Second, the current framework relies on offline fine-tuning, which necessitates prior knowledge

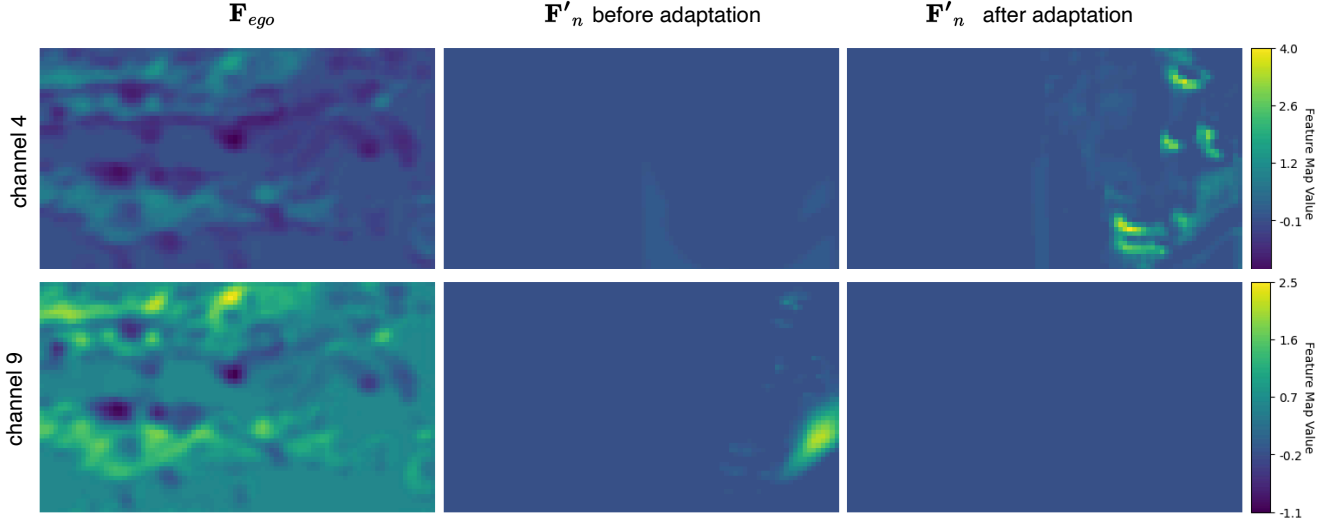


Fig. 7: Visualization of the feature maps of the L_p+L_p scenario. We show the feature map with a downsampling ratio of 4x.

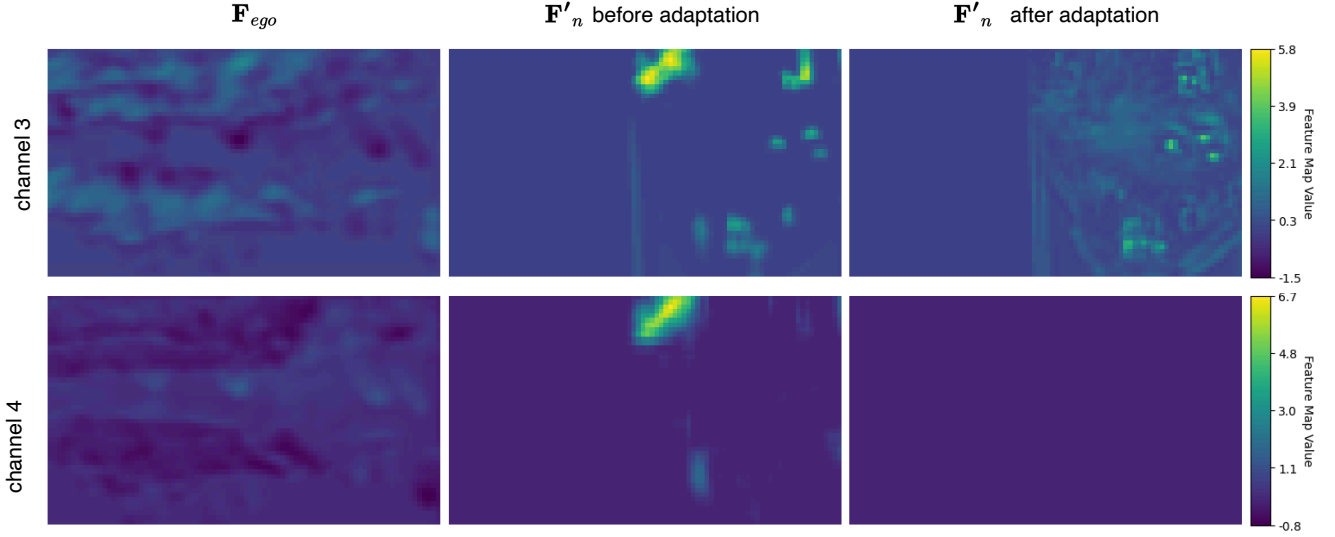


Fig. 8: Visualization of the feature maps of the $L_p+L_p^{seg}$ scenario. We show the feature map with a downsampling ratio of 4x.

of basic information during the fine-tuning stage. This dependency partially restricts the framework's adaptability in dynamic or unknown environments. Future work could explore the potential of online adaptation mechanisms to enhance the framework's flexibility and applicability in real-time scenarios.

VII. CONCLUSION

In conclusion, this paper introduces IMCP, the first intermediate model-agnostic cooperative perception framework. IMCP overcomes the limitations inherent in existing approaches that necessitate joint training. By incorporating hierarchical feature adaptation layers, IMCP facilitates robust and flexible cooperation among agents employing diverse models and training objectives while maintaining parameter efficiency. This approach

mitigates concerns regarding intellectual property and enables decentralized deployment. Evaluation on real-world datasets demonstrates that IMCP achieves competitive performance with joint training methods, establishing its potential as a practical and effective solution for real-world cooperative perception systems and advancing the field towards model-agnostic cooperative perception.

VIII. ACKNOWLEDGE

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